

Math Object and Number Methods & Date and Time in JavaScript



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01

Introduction

Introduction

The Math object provides built-in methods and properties for mathematical calculations.

JavaScript uses the Date object to handle dates, times, timestamps, etc.





02

Math Object

Common Math Methods & Properties

Method / Property	Description	Example
Math.PI	Value of π	Math.PI $\rightarrow$ 3.14159...
Math.E	Euler's number	Math.E $\rightarrow$ 2.71828...
Math.round(x)	Rounds to nearest integer	Math.round(4.6) $\rightarrow$ 5
Math.floor(x)	Rounds down	Math.floor(4.9) $\rightarrow$ 4
Math.ceil(x)	Rounds up	Math.ceil(4.1) $\rightarrow$ 5
Math.trunc(x)	Removes decimal part	Math.trunc(4.7) $\rightarrow$ 4
Math.abs(x)	Absolute value	Math.abs(-10) $\rightarrow$ 10
Math.pow(x, y)	x to the power of y	Math.pow(2, 3) $\rightarrow$ 8
Math.sqrt(x)	Square root	Math.sqrt(16) $\rightarrow$ 4
Math.min(...args)	Smallest number	Math.min(5, 1, 8) $\rightarrow$ 1
Math.max(...args)	Largest number	Math.max(5, 1, 8) $\rightarrow$ 8
Math.random()	Random float between 0 and 1	Math.random() $\rightarrow$ 0.56...





03

Number Methods

Common Math Methods & Properties

Number Conversion Methods

Method	Description	Example
Number()	Converts a value to number	Number("5") → 5
parseInt()	Parses integer from string	parseInt("42px") → 42
parseFloat()	Parses float from string	parseFloat("3.14text") → 3.14
toFixed(n)	Rounds to n decimal places (string)	3.14159.toFixed(2) → "3.14"
isNaN(x)	Checks if value is NaN	isNaN("abc") → true
isFinite(x)	Checks if value is finite	isFinite(10/0) → false





04

Date and Time

Create Date



Example :

JS index.js

JS index.js > ...

```
1 let date = new Date();  
2 console.log(date);    // 2025-07-09T07:49:24.559Z  
3  
4  
5  
6
```

Common Date Methods

Method	Description	Example Output
getFullYear()	Year (e.g., 2025)	2025
getMonth()	Month (0–11)	0 = January
getDate()	Day of the month (1–31)	15
getDay()	Day of week (0–6, Sun = 0)	2 = Tuesday
getHours()	Hour (0–23)	14
getMinutes()	Minutes	30
getSeconds()	Seconds	45
getMilliseconds()	Milliseconds	123
getTime()	Milliseconds since Jan 1, 1970	1.6999E+12
toString()	Date part as readable string	"Tue Jul 08 2025"
toTimeString()	Time part	"14:34:12 GMT+0530"
toISOString()	ISO format	"2025-07-08T09:04:23Z"



Set Methods

Method	Purpose	Example
setFullYear(y)	Change year	date.setFullYear(2030)
setMonth(m)	Set month (0–11)	date.setMonth(5)
setDate(d)	Set day of month	date.setDate(15)
setHours(h)	Set hour	date.setHours(10)
setMinutes(m)	Set minutes	date.setMinutes(45)

